

ForesightNav: Anticipatory Reinforcement Learning for UAV Navigation in Environments with Dynamic Obstacles and Large Non-Convex Structures

Abstract—Safe unmanned aerial vehicle navigation among dynamic obstacles and large non-convex structures requires anticipating motion conflicts while avoiding local trapping near large walls and concave geometry. However, many local learning-based navigation policies rely primarily on instantaneous geometry and distance-based safety signals, which can delay avoidance and favor locally safe but globally unproductive actions. This paper presents ForesightNav, an anticipatory reinforcement-learning framework combining finite-horizon motion-risk modeling with predictive outcome supervision. A time-to-collision-aware three-dimensional velocity-obstacle reward incorporates relative motion and obstacle geometry to encourage proactive dynamic-obstacle avoidance. An action-conditioned multi-horizon objective directly supervises future collision and local-trapping risks, minimum clearance, and goal progress through the shared actor-critic encoder. Both proposed learning mechanisms are confined to training, preserving direct policy inference at deployment. Extensive simulation experiments demonstrate that ForesightNav outperforms the compared state-of-the-art methods in dynamic-obstacle avoidance and navigation through environments containing large non-convex structures. Zero-shot physical flights in four environments further demonstrate deployment feasibility without fine-tuning. Code and trained models will be made publicly available upon publication.

I. INTRODUCTION

Autonomous navigation is essential for unmanned aerial vehicles (UAVs) operating in environments with dynamic obstacles and large non-convex structures. Beyond reacting to local geometry, a UAV must account for the likely consequences of its decisions: an action that is collision-free at the current instant may still create a future motion conflict or steer the UAV into a local deadlock near an extended wall or within a concavity. Providing such foresight under limited onboard sensing and computation remains a central challenge in UAV navigation and control.

Classical navigation systems decompose the problem into perception, prediction, planning, and control [1]–[3]. Although interpretable and constraint-aware, they often rely on accurate models, hand-crafted objectives, and repeated online optimization, which can increase tuning effort and online computation. Deep reinforcement learning offers an alternative by learning direct observation-to-action policies from large-scale simulated experience [4]–[6]. NavRL [6], for example, demonstrates the potential of this paradigm by combining Proximal Policy Optimization (PPO) [7], bounded three-dimensional velocity control, and efficient parallel training to produce responsive navigation commands through direct policy inference. However, its published ob-

jective relies on local distance and goal-progress signals and does not explicitly model finite-horizon relative-motion conflicts or directly supervise whether locally safe action sequences circumnavigate an extended obstacle or escape a concavity [6].

These limitations produce two recurring failure modes. First, instantaneous distance-based rewards can assign similar penalties to obstacles at similar distances, even though closing speed determines how quickly separation decreases and vertical alignment determines whether the three-dimensional safety volumes can overlap. These factors can yield substantially different times to collision, causing a rapidly approaching obstacle to elicit an adequate response only after the available avoidance window has narrowed. Second, locally collision-free actions can remain globally unproductive around large non-convex structures. Broad walls repeatedly obstruct the goal direction, while L- and U-shaped structures can trap a policy in oscillatory or stagnant behavior that appears safe at each individual step [8]. Successful circumnavigation instead requires a temporally consistent bypass whose immediate goal progress may be small or temporarily negative. These observations motivate predictive supervision during training without requiring online rollout evaluation at deployment.

To address these limitations, ForesightNav introduces two complementary training mechanisms. First, building on the classical velocity obstacle (VO) concept [9], it integrates finite-horizon three-dimensional VO (3D-VO) geometry with axis-specific safety margins and its earliest valid time to collision (TTC) into a bounded, TTC-graded risk reward for policy optimization. Second, an action-conditioned multi-horizon objective predicts future collision and local-trapping risks, minimum forward clearance, and cumulative goal progress. The reward shapes return-based policy updates, whereas the auxiliary losses train the observation encoder shared by the actor and critic. Both proposed learning mechanisms are confined to training, preserving direct policy inference at deployment. Fig. 1 summarizes the overall flow from perception and predictive policy learning to deployment-time control.

We evaluate ForesightNav in simulation and physical environments. In simulation, we compare it with NavRL [6] and PE-Planner [1] across static, dynamic, mixed, and large non-convex settings, and conduct controlled component ablations in the dynamic and large non-convex environments. ForesightNav achieves higher success rates in the dynamic

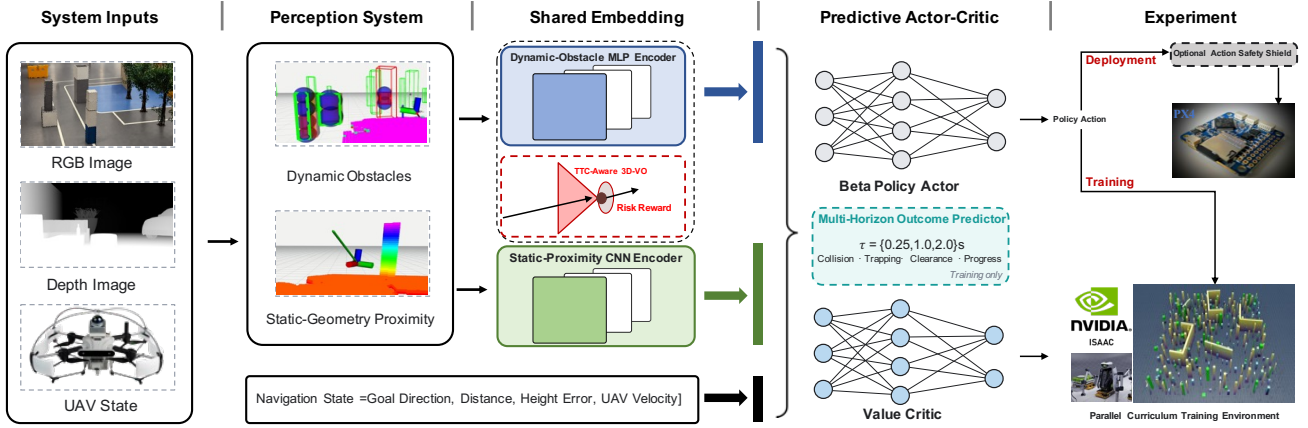


Fig. 1. The proposed ForesightNav framework. The perception system processes local perceptual observations together with the UAV state to form dynamic-obstacle, static-geometry proximity, and navigation representations. Modality-specific MLP and CNN encoders transform these inputs into a shared embedding for the predictive actor-critic. During training, the TTC-aware 3D-VO risk reward and action-conditioned multi-horizon outcome predictor jointly shape policy learning in parallel curriculum environments. During deployment, the outcome predictor is removed, and the Beta-policy actor directly generates velocity actions that can be further refined by an optional safety shield.

and mixed settings than the compared methods and, relative to NavRL and PE-Planner, fewer timeout and deadlock events around large non-convex structures. Zero-shot physical flights further demonstrate deployment feasibility without fine-tuning.

The main contributions of this work are summarized as follows:

- 1) **Physics-informed finite-horizon risk supervision:** We integrate 3D-VO geometry with axis-specific safety margins and earliest valid TTC into a bounded, TTC-graded training reward for proactive dynamic-obstacle avoidance.
- 2) **Action-conditioned predictive representation learning:** We introduce a multi-horizon auxiliary objective that directly supervises collision, local-trapping, clearance, and progress outcomes in the shared actor-critic encoder without deployment-time prediction.
- 3) **Simulation and physical evaluation:** Large-scale simulations demonstrate improved dynamic-obstacle avoidance and fewer timeout and deadlock events around large non-convex structures relative to external baselines, while zero-shot flights demonstrate deployment feasibility.

II. RELATED WORK

A. Classical, Learning-Based, and Hybrid Navigation

Classical UAV navigation commonly combines path search with receding-horizon trajectory optimization [1]–[3]. FASTER maintains dynamically feasible trajectories in unknown environments [2], ViGO optimizes vision-aided B-spline trajectories around dynamic obstacles [3], and PE-Planner couples kinodynamic search and trajectory refinement with model predictive contouring control [1]. These methods provide explicit feasibility constraints and planning lookahead, but depend on sufficiently accurate environment models and repeated online search or optimization, making

their performance sensitive to perception quality and computation budget. Learning-based methods instead shift much of this computation to offline experience: CADRL learns an interaction-aware value function [4], Long *et al.* map range and goal observations directly to continuous controls [5], and NavRL combines three-dimensional velocity control with a deployment-time safety shield [6]. Direct policy inference is efficient, but scalar rewards provide only indirect and entangled supervision of distinct future collision, local-trapping, clearance, and progress outcomes.

Large or non-convex structures expose a further distinction between immediate collision avoidance and successful obstacle circumnavigation. A global planner can reason about map connectivity and route around a large obstacle, but this capability requires adequate global coverage and repeated map updates and replanning, which are difficult to maintain in partially observed dynamic scenes. Local learning-based formulations avoid that dependency, yet act from local observations without explicitly encoding global route connectivity around extended structures [5, 6, 10]. They therefore provide limited guidance for maintaining a bypass direction along a broad wall or escaping an L- or U-shaped concavity, making locally safe deadlocks or local optima more likely. To address this issue, Lee *et al.* [8] use privileged time-of-arrival maps to guide quadrotors around large walls and dead ends.

Hybrid methods augment learned policies with online planning, recovery, or action projection [11]–[13], improving reachability or safety at the cost of additional modules and online computation. Training-time demonstrations can instead transfer structured guidance, but require suitable expert demonstration data [14]. These trade-offs motivate anticipatory policy learning with lightweight online inference.

B. Velocity-Obstacle-Guided Learning

The velocity-obstacle (VO) formulation identifies velocities that lead to collision under an assumed relative-motion model and supports online selection outside the resulting

forbidden set [9]. Optimal Reciprocal Collision Avoidance (ORCA) represents reciprocal multi-agent avoidance through pairwise linear half-plane constraints and computes a feasible velocity close to the preferred command [15]. Such methods provide local kinematic collision avoidance online, but when used only as action filters, they correct a nominal command after it has been generated and do not teach the policy to avoid hazardous actions proactively.

VO structure has also been incorporated into reinforcement-learning objectives. DRL-VO uses a planar VO-based heading reward with LiDAR histories and pedestrian kinematics [10], while Han *et al.* shape rewards using the Reciprocal Velocity Obstacle (RVO) formulation [16], specifically its area and expected collision time, for differential-drive robots [17]. NavRL instead applies a VO-inspired filter after policy inference [6]. Unlike these planar reward formulations or post-policy filtering alone, ForesightNav maps finite-horizon collision times from a three-dimensional VO model to a training reward. The resulting signal accounts for obstacle extent, vertical separation, and relative motion, while the runtime shield remains a separate safeguard for residual unsafe actions.

C. Predictive Models and Auxiliary Outcome Learning

Predictive navigation methods exploit future structure in different ways. NavRep studies unsupervised latent reconstruction and dynamics models for navigation policy learning [18], RGL combines relational interaction modeling with multi-step crowd lookahead [19], and BADGR predicts action-conditioned events to evaluate candidate control sequences online [20]. Their predictions support representation learning or explicit lookahead, rather than directly supervising a shared actor-critic representation with separate task outcomes at multiple horizons. Auxiliary objectives provide a complementary approach: general value functions predict temporally extended signals [21], UNREAL combines reward prediction with auxiliary control and value replay [22], and SPR enforces action-conditioned multi-step latent consistency [23]. These objectives improve learned representations, but do not directly provide the combined collision, local-trapping, clearance, and progress supervision required for the navigation failures considered here.

In contrast, ForesightNav jointly trains its shared policy representation using analytic finite-horizon motion risk and action-conditioned, horizon-specific navigation outcomes. This design targets late responses to dynamic motion conflicts and locally safe deadlocks around large non-convex structures, while the predictive branch is removed during deployment to preserve direct policy inference.

III. METHODOLOGY

In this section, we introduce the finite-horizon TTC-aware 3D-VO formulation, the reinforcement-learning problem, the predictive actor-critic, and the deployment-time shield. Here, “TTC-aware 3D-VO” denotes a training signal that maps the earliest valid collision within a finite-horizon

three-dimensional VO with axis-specific safety margins to bounded, TTC-graded risk, rather than an online planner.

A. Finite-Horizon TTC-Aware 3D Velocity-Obstacle Modeling

Instantaneous clearance cannot distinguish obstacles with similar separation but different relative motion. ForesightNav therefore extends the classical velocity-obstacle concept [9] to quantify finite-horizon 3-D motion conflicts. For the transition induced by the executed velocity command $\mathbf{a}_t^G$, the model uses the resulting UAV position $\mathbf{p}_{t+1}$ and realized velocity $\mathbf{v}_{t+1}$, expressed with the obstacle states in a common inertial frame, as illustrated in Fig. 2.

Obstacle i , with center $\mathbf{p}_{i,t}^o$, velocity $\mathbf{v}_{i,t}^o$, horizontal size (w_i^x, w_i^y) , and height ℓ_i^z , is represented by a centered vertical stack. Let $d_i = \max(w_i^x, w_i^y)$, $K_i = \max\{1, \lceil \ell_i^z/d_i \rceil\}$, and $b_i = \frac{1}{2}\sqrt{(w_i^x)^2 + (w_i^y)^2}$. Primitive $j = 0, \dots, K_i - 1$ has velocity $\mathbf{v}_{i,j,t}^o$ and center $\mathbf{p}_{i,j,t}^o = \mathbf{p}_{i,t}^o + [0, 0, (j - (K_i - 1)/2)d_i]^\top$, giving inter-center spacing d_i .

For horizon $H_{VO} > 0$, let $\mathbf{r}_{i,j,t+1} = \mathbf{p}_{i,j,t+1}^o - \mathbf{p}_{t+1}$ and $\mathbf{u}_{i,j,t+1} = \mathbf{v}_{i,j,t+1}^o - \mathbf{v}_{t+1}$. Writing r^z and u^z for their vertical components, the swept interval is $I_{i,j}^z = [\min\{r^z, r^z + H_{VO}u^z\}, \max\{r^z, r^z + H_{VO}u^z\}]$. We retain $(i, j) \in \mathcal{P}_{t+1}$ iff $I_{i,j}^z \cap [-b_i - m_{pf}, b_i + m_{pf}] \neq \emptyset$; hence, future vertical motion is considered before applying axis-specific inflation:

$$\begin{aligned} \mathbf{Q}_{i,j} &= \text{diag}(b_i + m_{xy}, b_i + m_{xy}, b_i + m_z), \\ \bar{\mathbf{r}}_{i,j,t+1} &= \mathbf{Q}_{i,j}^{-1} \mathbf{r}_{i,j,t+1}, \quad \bar{\mathbf{u}}_{i,j,t+1} = \mathbf{Q}_{i,j}^{-1} \mathbf{u}_{i,j,t+1}. \end{aligned} \quad (1)$$

Using $m_{pf} = 0.6 \text{ m}$ and $m_{xy} = m_z = 0.3 \text{ m}$, the inflated region is $\mathcal{E}_{i,j} = \{\boldsymbol{\xi} \in \mathbb{R}^3 : \|\mathbf{Q}_{i,j}^{-1} \boldsymbol{\xi}\|_2 \leq 1\}$. Under constant relative velocity, the finite-horizon velocity obstacle is

$$\mathcal{VO}_{i,j,t+1}^{H_{VO}} = \{\mathbf{v} \in \mathbb{R}^3 \mid \exists s \in (0, H_{VO}) : \|\mathbf{Q}_{i,j}^{-1} [\mathbf{r}_{i,j,t+1} + s(\mathbf{v}_{i,j,t+1}^o - \mathbf{v})]\|_2 \leq 1\}. \quad (2)$$

ForesightNav tests the realized velocity $\mathbf{v}_{t+1}$ for membership rather than constructing the full set. For each retained primitive, local coefficients A, B, C give

$$\begin{aligned} As^2 + Bs + C &= 0, \quad A = \|\bar{\mathbf{u}}_{i,j,t+1}\|_2^2, \\ B &= 2\bar{\mathbf{r}}_{i,j,t+1}^\top \bar{\mathbf{u}}_{i,j,t+1}, \\ C &= \|\bar{\mathbf{r}}_{i,j,t+1}\|_2^2 - 1. \end{aligned} \quad (3)$$

For a non-overlapping primitive, a valid time to collision exists when relative motion is nonzero and approaching, the discriminant $\Delta_{i,j,t+1}^{\text{quad}} = B^2 - 4AC$ is nonnegative, and

$$T_{i,j,t+1} = \frac{-B - \sqrt{\Delta_{i,j,t+1}^{\text{quad}}}}{2A} \in (0, H_{VO}]. \quad (4)$$

A primitive with $C \leq 0$ receives unit risk. Conflicts outside H_{VO} receive zero risk, while valid in-horizon conflicts are graded by an exponentially decaying function of TTC with decay constant $\tau_{VO} > 0$:

$$\rho_{i,j,t+1} = \begin{cases} 1, & C \leq 0, \\ \exp(-T_{i,j,t+1}/\tau_{VO}), & T_{i,j,t+1} \in (0, H_{VO}], \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

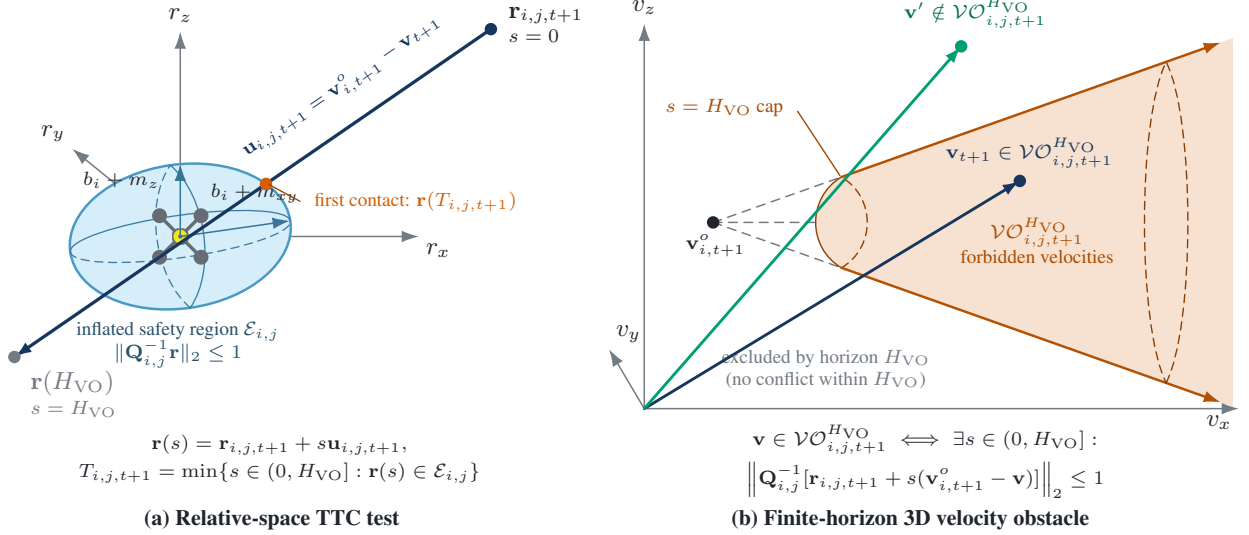


Fig. 2. Finite-horizon TTC-aware 3D-VO geometry for one retained primitive. (a) Relative motion first intersects the inflated region at $T_{i,j,t+1}$. (b) The corresponding realized velocity lies inside the finite-horizon VO; the $s = H_{VO}$ cap excludes conflicts beyond the prediction horizon.

Over the retained primitive pairs $\mathcal{P}_{t+1}$, define $\rho_{t+1}^{VO} = \max_{(i,j) \in \mathcal{P}_{t+1}} \rho_{i,j,t+1}$, with value zero when the set is empty. This bounded maximum selects the most imminent conflict without scaling with primitive count. It assumes constant relative velocity over the finite horizon and is not a formal safety guarantee.

B. Reinforcement-Learning Formulation

1) *Problem Formulation*: ForesightNav models local UAV navigation as a partially observable Markov decision process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \Omega, O, \gamma)$, with latent-state space $\mathcal{S}$, normalized-action space $\mathcal{A}$, transition P , reward R , observation space/model (Ω, O) , and discount γ . Given $\mathbf{o}_t \sim O(\cdot | \mathbf{x}_t)$, the policy samples $\tilde{\mathbf{a}}_t \sim \pi_{\theta, \omega}(\cdot | \mathbf{o}_t)$ to maximize $J(\theta, \omega) = \mathbb{E}_{\pi_{\theta, \omega}}[\sum_{t=0}^{T_{ep}-1} \gamma^t r_t]$. The hidden state $\mathbf{x}_t$ may supply rewards and auxiliary targets. The episode-fixed navigation frame G originates at the initial UAV position, with horizontal x -axis toward the goal and z -axis vertical.

2) *State and Observation Space*: A compact local observation compatible with the navigation interface in [6] combines static proximity, navigation state, and tracked dynamic obstacles:

$$\begin{aligned} \mathbf{o}_t &= (\mathbf{L}_t, \mathbf{s}_t, \mathbf{D}_t), \\ \mathbf{L}_t &\in \mathbb{R}^{36 \times 4}, \quad \mathbf{s}_t \in \mathbb{R}^8, \quad \mathbf{D}_t \in \mathbb{R}^{5 \times 10}, \\ \mathbf{s}_t &= [\hat{\mathbf{r}}_{g,t}^G, d_{g,t}^{xy}, \Delta z_{g,t}, \mathbf{v}_t^G], \\ \mathbf{d}_{i,t} &= [\hat{\mathbf{r}}_{i,t}^G, d_{i,t}^{xy}, \Delta z_{i,t}, \mathbf{v}_{i,t}^{o,G}, \mathbf{q}_{i,t}^{\text{size}}] \in \mathbb{R}^{10}. \end{aligned} \quad (6)$$

Here, $\mathbf{L}_t$ is a yaw-aligned proximity grid and $\mathbf{D}_t$ stacks up to five nearest in-range obstacle descriptors $\mathbf{d}_{i,t}$. The terms $\hat{\mathbf{r}}$, d^{xy} , and Δz denote UAV-to-target unit direction, horizontal separation, and target-minus-UAV altitude, respectively. Vectors are expressed in G , and $\mathbf{q}_{i,t}^{\text{size}} \in \mathbb{R}^2$ encodes obstacle size.

3) *Action Space*: Using a factorized Beta policy [6], the actor samples a normalized action and maps it to a componentwise-bounded three-dimensional velocity command in G :

$$\begin{aligned} \tilde{\mathbf{a}}_t &\sim \text{Beta}(\boldsymbol{\alpha}_t, \boldsymbol{\beta}_t), \\ \mathbf{a}_t^G &= v_{\text{axis}}(2\tilde{\mathbf{a}}_t - \mathbf{1}) \in [-v_{\text{axis}}, v_{\text{axis}}]^3. \end{aligned} \quad (7)$$

Here, $v_{\text{axis}} = 2.0 \text{ m/s}$ is a per-axis Cartesian command bound, not a bound on $\|\mathbf{a}_t^G\|_2$. The command is rotated into the world frame and tracked by a low-level controller; yaw is regulated separately.

C. Reward Function

The per-transition reward combines five shaping components with a terminal term:

$$r_t = \lambda_{VO} r_t^{VO} + \lambda_g r_t^{\text{goal}} + \lambda_{\text{sm}} r_t^{\text{sm}} + \lambda_{\text{ss}} r_t^{\text{ss}} + \lambda_{\text{alt}} r_t^{\text{alt}} + r_t^{\text{term}}. \quad (8)$$

For the mutually exclusive terminal labels, r_t^{term} equals R_{goal} , $-R_{\text{coll}}$, $-R_{\text{timeout}}$, or zero upon goal arrival, collision, timeout, or a nonterminal transition, respectively. In formula order, the shaping weights are (1.3, 35, 0.1, 0.6, 0.5), and $(R_{\text{goal}}, R_{\text{coll}}, R_{\text{timeout}}) = (50, 45, 45)$.

1) *TTC-Aware 3D-VO Risk Reward*: The finite-horizon risk in Sec. III-A enters the reward as

$$r_t^{VO} = -\eta(n) \rho_{t+1}^{VO}, \quad \eta(n) = \min\left(\frac{n}{N_{\text{warm}}}, 1\right), \quad (9)$$

where n is the optimization step and N_{warm} is the warmup duration in optimization steps. The warmup factor gradually introduces anticipatory collision-risk shaping and prevents it from dominating before basic goal-directed behavior is established.

2) *Base Shaping Rewards*: Let $d_t^g = \|\mathbf{p}_g - \mathbf{p}_t\|_2$, let $\ell_{k,t+1}^L$ be static-proximity ray range k , and write $[x]_+ = \max(x, 0)$. The remaining terms are

$$\begin{aligned} r_t^{\text{goal}} &= d_t^g - d_{t+1}^g, \\ r_t^{\text{sm}} &= -\|\mathbf{v}_{t+1} - \mathbf{v}_t\|_2, \\ r_t^{\text{ss}} &= -\frac{1}{N_L} \sum_{k=1}^{N_L} \left[1 - \frac{\ell_{k,t+1}^L}{m_s} \right]_+, \\ r_t^{\text{alt}} &= -[|z_{t+1} - z_g| - \epsilon_z]_+^2. \end{aligned} \quad (10)$$

Here, m_s is the static-obstacle margin and ϵ_z the altitude tolerance. These terms reward signed goal progress while discouraging abrupt velocity changes, static proximity, and unnecessary vertical detours.

D. Network Design and Policy Training

Following the network organization of NavRL [6], static-proximity and dynamic-obstacle encoders fuse their outputs with $\mathbf{s}_t$ into $\mathbf{z}_t = E_\omega(\mathbf{o}_t)$. The Beta actor $\pi_\theta(\cdot | \mathbf{z}_t)$, critic $V_\psi(\mathbf{z}_t)$, and outcome decoder share this observation-conditioned representation; none receives the latent state.

At simulator interval $\Delta t = 0.016$ s, we use physical horizons $\mathcal{T} = \{0.25, 1.0, 2.0\}$ s and $h_\tau = \text{round}(\tau/\Delta t) \in \{16, 63, 125\}$ transitions. A lightweight decoder predicts the action-conditioned outcomes at each $\tau \in \mathcal{T}$:

$$\hat{\mathbf{y}}_t^{(\tau)} = g_\phi^{(\tau)}(\mathbf{z}_t, \tilde{\mathbf{a}}_t) = \left[\hat{c}_t^{(\tau)}, \hat{\chi}_{\text{trap},t}^{(\tau)}, \hat{f}_t^{(\tau)}, \hat{g}_t^{(\tau)} \right], \quad (11)$$

where the outputs represent collision risk, local-trapping risk, minimum forward clearance, and cumulative goal progress. Let $W_t^{(\tau)} = \min(h_\tau, T_t^{\text{rem}})$ be the valid future-window length before episode termination. The rollout targets are

$$\begin{aligned} c_t^{(\tau)} &= \max_{1 \leq \kappa \leq W_t^{(\tau)}} c_{t+\kappa}, & f_t^{(\tau)} &= \min_{1 \leq \kappa \leq W_t^{(\tau)}} f_{t+\kappa}, \\ g_t^{(\tau)} &= \sum_{\kappa=1}^{W_t^{(\tau)}} \Delta d_{t+\kappa}^g. \end{aligned} \quad (12)$$

Here, c_t is the collision indicator, f_t is forward clearance, and $\Delta d_t^g = d_{t-1}^g - d_t^g$. For $b_t = \mathbb{I}[f_t \leq d_b]$, let $\bar{b}_t^{(\tau)}$, $\bar{v}_t^{(\tau)}$, and $\bar{q}_t^{(\tau)}$ denote the window-averaged blockage, net horizontal-displacement rate, and goal-progress rate, respectively. The forward-looking target is

$$\chi_{\text{trap},t}^{(\tau)} = \mathbb{I} \left[\bar{b}_t^{(\tau)} \geq \rho_b \wedge \bar{v}_t^{(\tau)} \leq v_{\text{net}} \wedge \bar{q}_t^{(\tau)} \leq v_{\text{prog}} \right], \quad (13)$$

Specifically, with $W = W_t^{(\tau)}$, $\bar{b}_t^{(\tau)} = W^{-1} \sum_{\kappa=1}^W b_{t+\kappa}$, $\bar{v}_t^{(\tau)} = \|\mathbf{p}_{t+W}^{xy} - \mathbf{p}_t^{xy}\|_2 / (W\Delta t)$, and $\bar{q}_t^{(\tau)} = g_t^{(\tau)} / (W\Delta t)$. We set $(d_b, \rho_b, v_{\text{net}}, v_{\text{prog}}) = (1.5 \text{ m}, 0.8, 0.15 \text{ m/s}, 0.10 \text{ m/s})$. Their conjunction detects stagnation without marking a productive lateral detour as trapped. Subsequent actions follow the rollout policy; only $\tilde{\mathbf{a}}_t$ is explicitly conditioned upon.

Because $h_{2.0s} = 125$ exceeds the 32-step PPO rollout, per-environment auxiliary sequence buffers carry transitions across PPO rollout boundaries; target construction is episode-aware and never crosses a reset. Using $\text{symlog}(x) =$

$\text{sign}(x) \log(1 + |x|)$, let $\tilde{f}_t^{(\tau)} = \text{symlog}(f_t^{(\tau)})$ and $\tilde{g}_t^{(\tau)} = \text{symlog}(g_t^{(\tau)})$. The multi-task loss is

$$\begin{aligned} \mathcal{L}_{\text{MH}} &= \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} [\ell_{\text{BCE}}(\hat{c}_t^{(\tau)}, c_t^{(\tau)}) + \ell_{\text{BCE}}(\hat{\chi}_{\text{trap},t}^{(\tau)}, \chi_{\text{trap},t}^{(\tau)}) \\ &\quad + \sum_{u \in \{f,g\}} \ell_{\text{SL1}}(\hat{u}_t^{(\tau)}, \tilde{u}_t^{(\tau)})]. \end{aligned} \quad (14)$$

Training minimizes $\mathcal{L} = \mathcal{L}_{\text{PPO}} + \lambda_{\text{MH}} \mathcal{L}_{\text{MH}}$. With actions and targets fixed, $\mathcal{L}_{\text{MH}}$ updates the decoder and shared encoder, not the actor or critic heads. The decoder is removed at deployment.

E. Deployment-Time Policy Action Safety Shield

An optional VO-based action shield [6] acts after policy inference, passing commands that satisfy its collision-avoidance and control constraints and minimally adjusting unsafe ones. It is excluded from optimization and complements the training-time TTC-aware 3D-VO objective: the policy learns anticipatory avoidance, while the shield handles residual unsafe actions.

IV. SIMULATIONS AND EXPERIMENTS

A. Training Details and Setup

The policy is trained in NVIDIA Isaac Sim using PyTorch in a $40 \text{ m} \times 40 \text{ m}$ environment with geometry up to 4.5 m high. Each Cartesian velocity-command component is bounded by 2.0 m/s. Dynamic-obstacle radii are $\{0.125, 0.25, 0.375, 0.50\}$ m and their speeds are sampled from 0.5 to 1.5 m/s. Training uses 1,024 parallel UAVs with randomized tasks on an NVIDIA GeForce RTX 4090 GPU; each curriculum stage takes approximately 12 hours. Fig. 3 shows the curriculum progression, and Table I lists the training hyperparameters.

For quantitative simulation evaluation, each method is tested in 1,000 matched trials per scenario under identical environment configurations and metric thresholds. Obstacles follow the scenario-specific motion process, and trials terminate upon goal arrival, collision, or the fixed time limit $T_{\text{max}} = 30$ s. Let $N_{\text{trial}} = 1,000$; N_{succ} , N_{coll} , and N_{timeout} denote the three terminal counts, while $N_{\text{dead}}^{\text{evt}}$ and $N_{\text{rec}}^{\text{evt}}$ denote detected and recovered deadlock events. We report:

- *Success Rate* ($N_{\text{succ}}/N_{\text{trial}}$): Goal reached before collision or timeout.
- *Collision Rate* ($N_{\text{coll}}/N_{\text{trial}}$): Termination by obstacle contact.
- *Timeout Rate* ($N_{\text{timeout}}/N_{\text{trial}}$): No goal arrival or collision within T_{max} .
- *Deadlock Event Count* ($N_{\text{dead}}^{\text{evt}}$): Transitions into deadlock after a forward obstacle and goal progress no greater than 0.005 m persist for 40 cycles (2 s at 20 Hz). The detector resets when either condition ceases, making each continuous segment one event. This evaluation-only event detector is distinct from the forward-looking local-trapping target used for auxiliary training.
- *Deadlock-Event Recovery Rate* ($N_{\text{rec}}^{\text{evt}}/N_{\text{dead}}^{\text{evt}}$): Fraction of events that, within 180 cycles (9 s), leave blockage, move

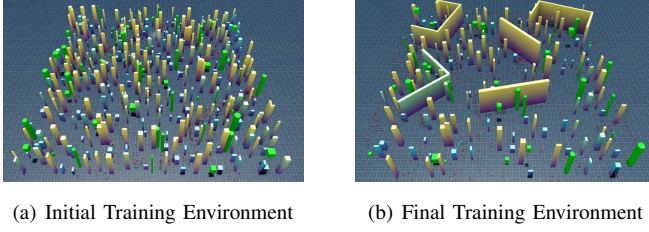


Fig. 3. Curriculum training in a $40\text{ m} \times 40\text{ m}$ environment. Dynamic-obstacle density increases progressively. (a) Initial environment with 60 dynamic obstacles. (b) Final environment with multiple large non-convex structures.

TABLE I
KEY HYPERPARAMETERS USED FOR POLICY TRAINING

Parameter	Value	Parameter	Value
Parallel UAVs	1,024	Rollout length	32
PPO epochs	4	Minibatches	16
Discount factor γ	0.99	GAE parameter λ_{GAE}	0.95
Initial LRs (en-coder/actor/critic)	$(5, 5, 5) \times 10^{-4}$	Adaptation LRs (en-coder/actor/critic)	$10^{-5}, 10^{-5}, 10^{-4}$
Outcome horizons $\mathcal{T}$	0.25/1.0/2.0 s	Outcome-loss weight λ_{MH}	0.1
VO horizon H_{VO}	(16/63/125)	TTC decay τ_{VO}	0.75 s
VO reward weight λ_{VO}	2.1 s	VO warmup	20,000 steps
	1.3		

0.3 m horizontally, and either reduce goal distance by 0.2 m or increase forward clearance by 0.2 m; undefined if no event occurs.

B. Simulation Results

We evaluate sim-to-sim transfer in Gazebo across static, dynamic, mixed, and large non-convex settings. Each matched trial samples the start on one environment boundary and the goal on the opposite boundary. We compare against NavRL [6] and PE-Planner [1]. Backbone removes both proposed mechanisms; ForesightNav w/o MH and ForesightNav w/o 3D-VO remove the corresponding training objective. PE-Planner retains its native pipeline, whereas all learned methods use the same deployment-time shield. Fig. 4 shows representative ForesightNav trajectories, Fig. 5 shows the corridor test, and Table II reports the quantitative results.

ForesightNav remains comparable to NavRL in the static environment and attains the highest success and lowest collision rates among the complete methods in the dynamic and mixed settings. In the dynamic ablation, adding the TTC-aware 3D-VO reward in the two matched pairs, ForesightNav w/o MH versus Backbone and ForesightNav versus ForesightNav w/o 3D-VO, increases success by 13.3 and 13.6 percentage points, respectively, with equal reductions in collision rate. The full method also improves success by 13.4 points over NavRL in the dynamic setting and by 14.7 points in the mixed setting. The smaller gains from adding MH indicate that motion-risk supervision is the dominant contributor to success and collision performance in these

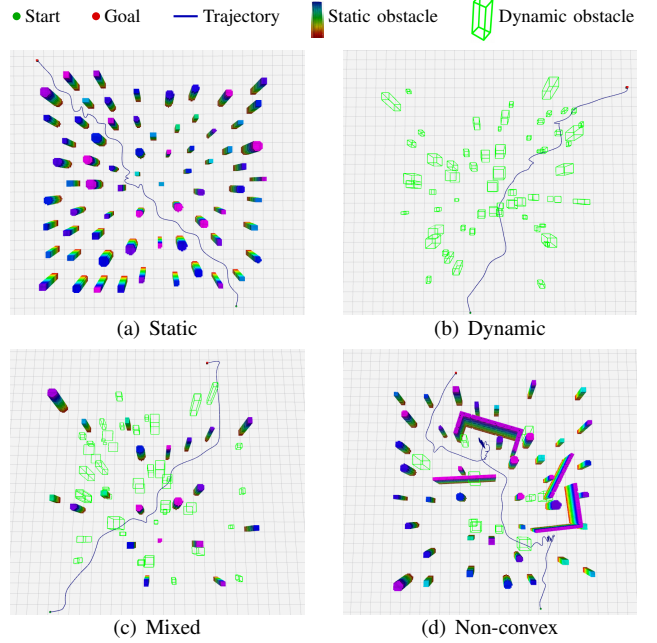


Fig. 4. Representative ForesightNav trajectories tested in four $20\text{ m} \times 20\text{ m}$ simulation environments.

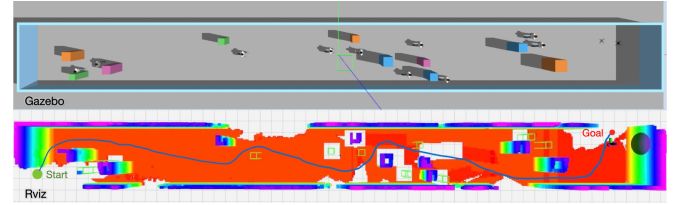


Fig. 5. ForesightNav navigation in a simulated corridor environment. The upper panel shows the Gazebo scene containing static obstacles and moving pedestrians, while the lower panel shows the corresponding environment map and UAV trajectory.

motion-intensive environments.

Around large non-convex structures, adding multi-horizon learning to Backbone raises success from 64.5% to 75.9%, reduces timeouts from 20.2% to 10.4%, and reduces deadlock events from 102 to 72. Adding it to ForesightNav w/o MH similarly raises success from 68.0% to 78.1%, reduces timeouts from 16.9% to 6.3%, and reduces deadlock events from 113 to 53. Although NavRL has the numerically lowest collision rate and ForesightNav w/o 3D-VO the numerically highest recovery rate, the full method attains the highest success and the fewest unresolved stalls. This pattern supports the primary role of multi-horizon supervision in sustained detours rather than uniformly lower collision rates.

To isolate the contribution of deployment-time action correction, Table III compares NavRL and ForesightNav with and without the safety shield in the purely dynamic-obstacle environment.

The shield increases success and reduces collisions for both methods, with a larger gain for ForesightNav. This supports the complementarity of anticipatory policy learning and residual online correction. Unshielded ForesightNav also

TABLE II

SYSTEM-LEVEL COMPARISON AND COMPONENT ABLATIONS IN FOUR $20\text{ m} \times 20\text{ m}$ ENVIRONMENTS (1,000 TRIALS PER METHOD AND ENVIRONMENT). PE-PLANNER USES ITS NATIVE PIPELINE; LEARNED METHODS USE A COMMON SHIELD. BOLD MARKS THE BEST NON-TIED RESULT; N/A DENOTES UNDEFINED RECOVERY WHEN NO DEADLOCK EVENT OCCURS.

Environment	Method	Success Rate (%) $\uparrow$	Collision Rate (%) $\downarrow$	Timeout Rate (%) $\downarrow$	Deadlock Events / 1,000 Trials $\downarrow$	Deadlock-Event Recovery Rate (%) $\uparrow$
Static obstacles ($N_{\text{stat}} = 70$)	PE-Planner	85.5	14.5	0	0	N/A
	NavRL	94.1	5.9	0	0	N/A
	ForesightNav	94.4	5.6	0	0	N/A
Dynamic obstacles ($N_{\text{dyn}} = 60$)	PE-Planner	65.6	34.4	0	0	N/A
	NavRL	69.9	30.1	0	0	N/A
	Backbone	68.3	31.7	0	0	N/A
	ForesightNav w/o MH	81.6	18.4	0	0	N/A
	ForesightNav w/o 3D-VO	69.7	30.3	0	0	N/A
	ForesightNav	83.3	16.7	0	0	N/A
Mixed static and dynamic obstacles ($N_{\text{stat}} = 20, N_{\text{dyn}} = 40$)	PE-Planner	63.2	36.8	0	0	N/A
	NavRL	61.8	38.2	0	0	N/A
	ForesightNav	76.5	23.5	0	0	N/A
Static obstacles and large non-convex structures	PE-Planner	58.9	15.9	25.2	127	21/127 (16.5%)
	NavRL	66.7	13.6	19.7	93	23/93 (24.7%)
	Backbone	64.5	15.3	20.2	102	27/102 (26.5%)
	ForesightNav w/o MH	68.0	15.1	16.9	113	29/113 (25.7%)
	ForesightNav w/o 3D-VO	75.9	13.7	10.4	72	33/72 (45.8%)
	ForesightNav	78.1	15.6	6.3	53	24/53 (45.3%)

TABLE III

EFFECT OF THE DEPLOYMENT-TIME SAFETY SHIELD OVER 1,000 TRIALS IN A $20\text{ m} \times 20\text{ m}$ DYNAMIC-OBSTACLE ENVIRONMENT WITH $N_{\text{dyn}} = 60$.

Method	Shield	Success Rate (%) $\uparrow$	Collision Rate (%) $\downarrow$	Timeout Rate (%) $\downarrow$
NavRL	Off	63.5	36.5	0
	On	69.9	30.1	0
ForesightNav	Off	71.4	28.6	0
	On	83.3	16.7	0

exceeds shielded NavRL in success (71.4% versus 69.9%), indicating that the gain does not arise solely from the shield.

C. Physical Flight Tests

To evaluate zero-shot sim-to-real transfer, we deploy the policy selected in simulation on a quadrotor without additional fine-tuning. The physical environments and corresponding velocity profiles are summarized in Fig. 6. The experiments are conducted in a $5\text{ m} \times 15\text{ m}$ area, and the norm of the commanded translational velocity is limited to 1.5 m/s . The onboard navigation stack supplies the policy with the same local observation schema used during training, comprising proximity measurements, the estimated UAV navigation state, and tracked dynamic-obstacle de-

scriptors. ForesightNav produces three-dimensional velocity commands that are tracked by the low-level flight controller. The multi-horizon outcome predictor and TTC-aware 3D-VO risk computation are used only for training and are not evaluated onboard.

Fig. 6(a)–(d) evaluate static clearance, response to obstacle motion, mixed constraints, and sustained detouring around non-convex structures. All tests use the same policy and runtime parameters without scene-specific tuning; the last requires lateral progress while direct goal motion remains blocked. Fig. 6(e) shows repeated acceleration and deceleration as the vehicle alternates between goal-directed motion and local avoidance under the common 1.5 m/s limit. Together, the flights and velocity histories provide qualitative evidence of closed-loop deployment under real sensing and control effects; controlled statistical comparisons remain those reported in simulation.

V. CONCLUSION

This paper presented ForesightNav for UAV navigation among dynamic obstacles and large non-convex structures. Its finite-horizon TTC-aware 3D-VO reward captures relative-motion risk, while action-conditioned multi-horizon supervision makes the shared representation predictive without adding deployment computation. Simulations demonstrate improved dynamic avoidance and robustness to

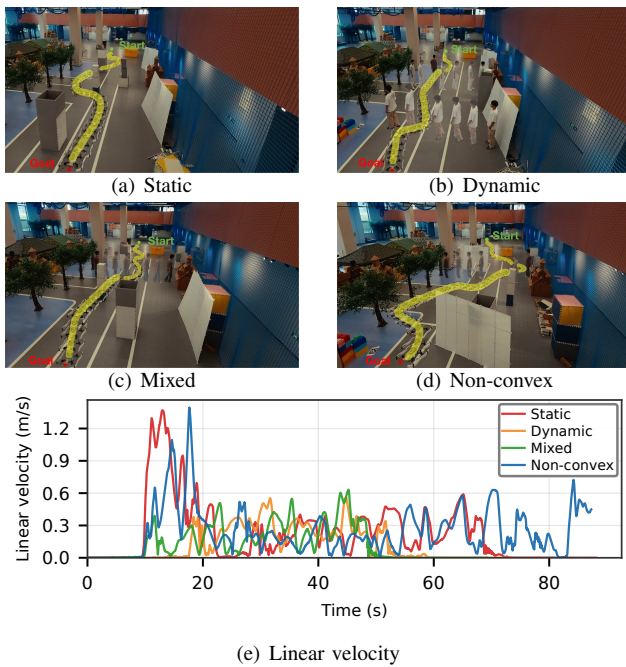


Fig. 6. Zero-shot ForesightNav flight tests in a 5 m $\times$ 15 m area: (a) static obstacles, (b) dynamic obstacles, (c) mixed obstacles, and (d) large non-convex structures. (e) UAV linear-velocity profiles in the four environments.

timeouts and local trapping, and zero-shot flights support deployment feasibility. These findings indicate that task-relevant predictive supervision can improve both short-term collision response and temporally consistent detours without transferring the auxiliary predictor to the vehicle. The method remains limited by local observations, a constant-relative-velocity assumption, and the lack of formal safety guarantees. Future work will extend ForesightNav to co-operative multi-UAV navigation in complex environments, emphasizing decentralized coordination, inter-UAV collision avoidance, and robust decision making under partial observability and limited communication.

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